HP Remote Device Access

Service Brief

Contents

Reference number: 5900-1110

Fourth edition: June 2013

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What is Remote Device Access?

Remote Device Access (RDA) is an IT solution that allows HP Support Specialists to connect securely from the HP network to systems on a customer's network for the purpose of problem diagnosis and troubleshooting as well as proactive support activities.

Key Features and Benefits

RDA allows HP to support its customers without a direct visit to the customer's sites. This results in quicker problem resolution for customers and reduced cost for HP. Once logged onto the customer's system, the HP Support Specialist provides reactive and/or proactive remote hardware and software support. With RDA HP Support Specialists can:

- Obtain additional information from the customer's monitored client
- Implement changes to the target system
- Perform proactive services
- Transfer files from the HP Support Specialist's desktop to the customer's monitored client
- Transfer files from the customer's monitored client to the HP Support Specialist's desktop

Remote Device Access Components

The RDA solution comprises a number of separate components, some of which are deployed on the HP network and some of which are deployed on the customer network.

- **Remote Access Portal System**

  The Remote Access Portal System (RAPS) is a server and web application located on the HP intranet that allows authorized HP Support Specialists to make secure connections from their PC within HP to remote customer systems.

- **Remote Connectivity Toolbox System**

  The Remote Connectivity Toolbox System (RCTS) is a web portal located on the HP intranet that allows authorized HP personnel to order remote connectivity solutions and manage customer connectivity information.

- **Remote Connectivity Database**

  All RDA data is stored in a database named the Remote Connectivity Database (RCDB).

- **Remote Access Connectivity Server**

  The Remote Access Connectivity Server (RACS) is an SSH server located on the HP network used to establish connections from HP to customer networks for troubleshooting and support purposes.

- **Remote Access Repository System**

  The Remote Access Repository System (RARS) is an Internet-facing and intranet-facing system used to hold package repositories for the Virtual CAS, and to hold router configurations for router recovery operations.

- **Remote Access Midway Server**

  The Remote Access Midway Server (RAMS) is a server that provides the Meet In The Middle connection paradigm which is a new approach to RDA connectivity. The salient feature of this approach is that a connection is made from the HP network to a customer network by meeting an inbound connection from the customer "in the middle". The primary function of the RAMS is to ensure that HP support agents and customers using the same connection key are matched up and connected together.

- **Customer Access System**

  The Customer Access System (CAS) is a customer-designated entry point into the customer network. The basic CAS is a system in the customer's network that HP can reach with SSH.

HP uses the CAS to tunnel connections (via SSH port forwarding) to target systems in the customer's network. No special setup is required, other than that the system is running an SSH daemon.

- **Virtual Customer Access System**

  The Virtual CAS (vCAS) is a pre-packaged CAS that includes advanced authentication, access control, and auditing capabilities. Customers may download and run this virtual appliance on a VMware ESX Server or VirtualBox host system instead of a basic CAS.

- **HP Instant Customer Access Server**

  The HP Instant CAS (HP iCAS) manages the customer's side of the Meet In The Middle connection. It is controlled by the iCAS start page on the RAMS. Its two main features are that it allows the customer to connect outbound via SSH to a RAMS on the HP network and also allows the customer to control access to targets on their network.

The Customer Access System

A Customer Access System (CAS) is required for all RDA connection methods, except Virtual Support Room. The CAS hosts the SSH server and provides a central point for customers to control remote access into their environment. Customers determine the login of each HP user individually to allow or deny specific services or access to specific computers within their network. HP uses the CAS to tunnel connections (via SSH port forwarding) to target systems in the customer's network. No special setup is required, other than that the system is running an SSH daemon.

The CAS can be set up as a virtual system using the RDA Virtual CAS, which is a virtual appliance that runs within the VMware ESX Server or VirtualBox environment. The Virtual CAS is a pre-packaged CAS that includes advanced authentication, access control, and auditing capabilities. Customers may download and run this virtual appliance on a VMware ESX Server or VirtualBox host system, instead of a basic CAS.

The Virtual CAS kit is around 113 MB and has the following key characteristics:

- **Authentication**

  The Virtual CAS provides a single sign on authentication mechanism. When an HP user logs into a virtual CAS as they connect to target systems, the CAS authentication occurs automatically.

- **Access Control**

  From the Web user interface the customer may control which HP support agents can access the CAS and restrict to where they can connect.

- **Audit Logs**

  The Virtual CAS maintains a detailed log, visible to the customer, of remote access sessions through the CAS into the customer environment. The log details the HP support agent’s email address, the date and time of the RDA session and the details of the target the HP support agent connected to.

- **Manageability**

  The Virtual CAS has an integrated patch and update mechanism. The customer may select to have patches and updates applied automatically, or they may manually apply them.

- **Low Footprint/Low Cost**

  The Virtual CAS is small in size (<115MB) and does not rely on licensed third party software.

The CAS can also be setup instantly using the HP Instant CAS, which is installed on the customer's desktop. The HP iCAS is used by a HP customer to initiate and manage connections from a point in their network to the HP RAMS server in order to "meet" the HP Service Engineer for a remote support session.

The Instant CAS kit is around 1.40 MB and has the following capabilities:

- Provides a browser base user interface
- Provides a mechanism for connecting to HP

- Provides a mechanism for connecting to HP through a proxy
- Provides a mechanism for viewing which systems (targets) the HP SE connects to
- Provides a mechanism for forcing a disconnect from a given target system
- Provides a simple white Access Control List (ACL) for managing to which targets an HP SE may connect
- Provides a session logging mechanism
- Provides a way to download the software from a central HP site for either an initial install or an upgrade

Remote Device Access Security

Remote Device Access requires a connection from HP to a customer-designated access server. HP understands that IT security policies within organizations vary considerably. Therefore, HP offers a number of remote access solutions (depending on the service level agreement) that help meet customer's security requirements. All of HP solutions use standard techniques that include SSH, IPsec, and HTTPS.

All attended RDA connection attempts from HP to customers are logged. The acting user, start and stopping times of the connection, and the connection status are logged. The connection status will indicate failures such as improper authentication and authorization.

For more information about Remote Device Access security, see the *HP Remote Device Access Security Overview* available at **http://www.hp.com/go/rda-docs**.

Remote Device Access Connectivity

At its simplest, an RDA connection to a customer involves the HP Support Specialist making the connection, and a target device in the customer's network. The specialist launches an application client on their desktop (such as VNC, RDC, a file transfer program, etc).

A connection to a customer involves a few more systems. At the HP side, connections must go through an Access Server. This is done so HP can authenticate the user, and verify that they are authorized to connect to the customer. The HP access server also records basic information about the connection (who, when, what, where, etc) for our audit records.

Presently there are several methods used for Remote Device Access:

- **SSH-Direct**

  The SSH protocol is used across the public Internet to and from HP for the connection.
- **hpVPN- Note: This is no longer offered in any region.**

  HP supplies the customer or partner with a small VPN router. This router is used to establish a secure connection to and from HP. This is the most secure connection option, which uses IPsec and SSH tunnels.
- **CorVPN**

  This is similar to the hpVPN solution, except that the customer-side router (or partner-side) is owned by the customer. The "Cor" stands for Customer Owned Router.
- **SSL VPN**

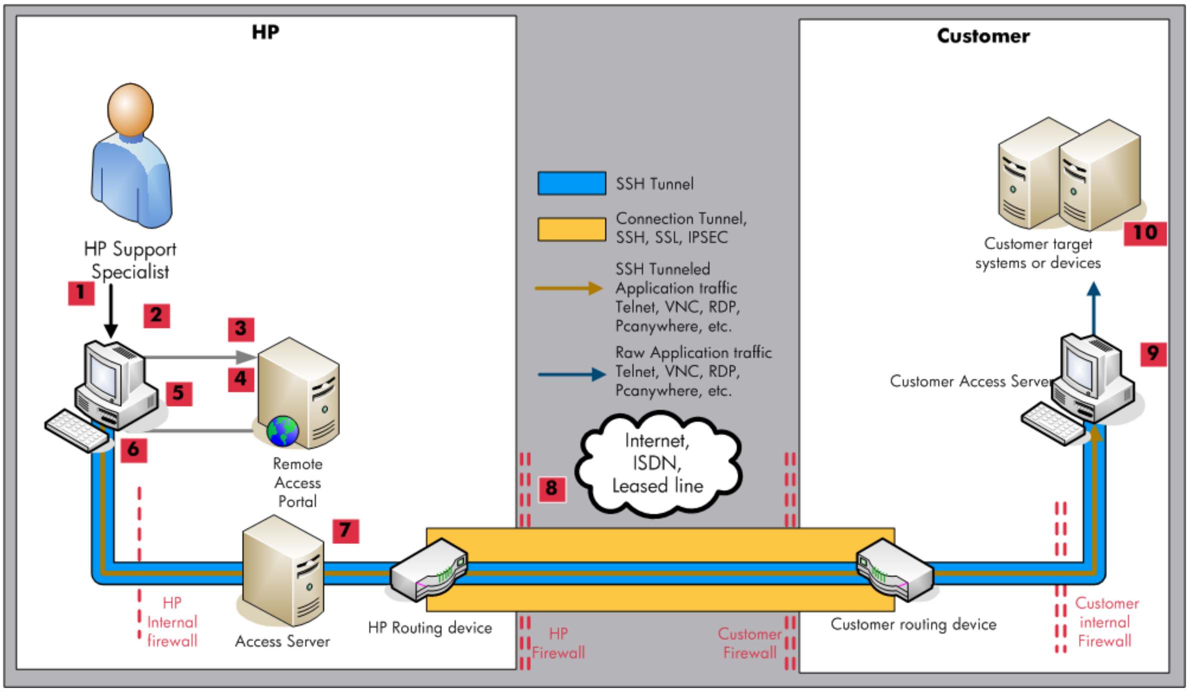
  The customer supplies and maintains an SSL VPN appliance. They then supply HP with the information on how to access this appliance, and their customer network beyond. The SSL protocol is used for communications.
- **Virtual Support Room (VSR)**

  This is an ad-hoc solution using HP Virtual Rooms.

Remote Device Access Customer Connection

The following shows the steps necessary to connect from HP to a customer's environment using the RDA solution.

Figure 1 Connection Steps



1. The HP Support Specialist logs onto the Remote Access Portal using a Class A DigitalBadge. (HP-issued X.509 certificates, internally called Class A DigitalBadges, that employ two-factor authentication.)
2. The HP Support Specialist selects a customer in the RAP. The specialist is authorized for selected customers.
3. The RAP retrieves customer connection information.
4. The HP Support Specialist selects the appropriate remote access application, for example, VNC, Terminal Services, etc.
5. The RAP generates an SSH profile and returns the profile to the HP Support Specialist's PC.
6. The HP Support Specialist opens the SSH profile invoking an SSH connection from the specialist's PC to the regional HP Access Server. The specialist may also invoke debugging tools on or off his desktop.
7. The HP Remote Access Server checks the HP Support Specialist's Class A DigitalBadge against the certification revocation list and also verifies that the HP Support Specialist is authorized to connect to the customer's environment.
8. A connection is then established between HP and the customer's environment, using Internet SSH, SSL VPN, or IPsec VPN, depending on which RDA solution is configured.
9. The HP Support Specialist authenticates on the Virtual Customer Access System (vCAS) automatically using a Class A DigitalBadge. The customer will have pre-approved this authentication or can authorize this access on the fly. The customer controls who can connect and what systems they are allowed to connect to.
10. The HP Support Specialist authenticates on the target support system by entering the logon and password credentials provided by the customer.
11. Once the connection is made, the application is automatically launched on the HP Support Specialist's desktop and the specialist can provide support on the customer's system.